

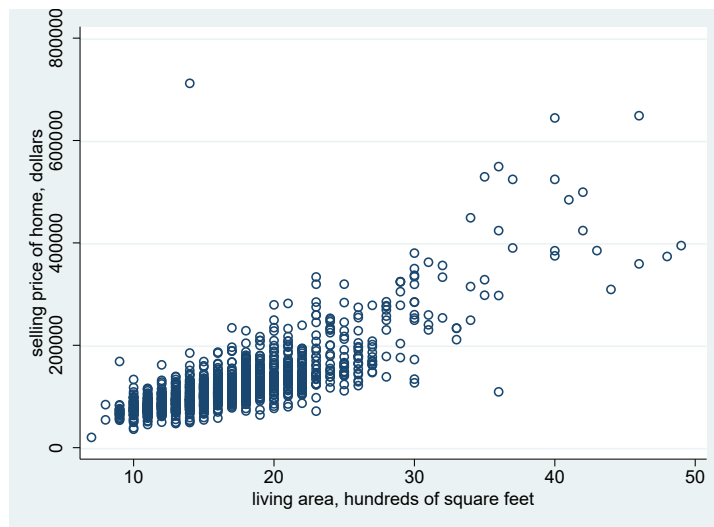
Stata Lab Session

How to Answer Empirical Questions in Homework

- 2.12* The file *stockton4.dat* contains data on 1500⁹ houses sold in Stockton, CA during 1996–1998. Variable descriptions are in the file *stockton4.def*.
- (a) Plot house selling price against house living area for all houses in the sample.
 - (b) Estimate the regression model $SPRICE = \beta_1 + \beta_2 LIVAREA + e$ for all the houses in the sample. Interpret the estimates. Draw a sketch of the fitted line.
 - (c) Estimate the quadratic model $SPRICE = \alpha_1 + \alpha_2 LIVAREA^2 + e$ for all the houses in the sample. What is the marginal effect of an additional 100 square feet of living area for a home with 1500 square feet of living area?
 - (d) In the same graph, plot the fitted lines from the linear and quadratic models. Which seems to fit the data better? Compare the sum of squared residuals (*SSE*) for the two models. Which is smaller?
 - (e) Estimate the regression model in (c) using only houses that are on large lots. Repeat the estimation for houses that are not on large lots. Interpret the estimates. How do the estimates compare?
 - (f) Plot house selling price against *AGE*. Estimate the linear model $SPRICE = \delta_1 + \delta_2 AGE + e$. Interpret the estimated coefficients. Repeat this exercise using the log-linear model $\ln(SPICE) = \theta_1 + \theta_2 AGE + e$. Based on the plots and visual fit of the estimated regression lines, which of these two models would you prefer? Explain.
 - (g) Estimate a linear regression $SPRICE = \eta_1 + \eta_2 LGELOT + e$ with dependent variable *SPRICE* and independent variable the indicator *LGELOT* which identifies houses on larger lots. Interpret these results.

(Obviously I don't expect the exact wording in your answers.)

- (a) By using the command “`twoway (scatter sprice livarea, msymbol(Oh))`” we are able to obtain the following graph. There is an obvious positive relationship between selling price and living area, though the relationship appears to be “curved” a little.



- (b) We obtain the following regression results: $\widehat{sprice} = -30069.2 + 9181.7livarea$. The interpretations are: if living area is zero, the selling price would be -\$30,069 (this doesn't make much sense as we explained in class); for every one hundred square feet of living area (note how livarea is defined as # of hundred sq ft), the selling price goes up by \$9,181.7. The fitted line can be seen in the graph below (or you can draw one by hand).



- (c) We obtain the following regression results: $\widehat{sprice} = 57728.3 + 212.6 livarea^2$. The marginal effect of 100 sq ft for a 1500 sq ft house is (note that variables are measured in 100 sq ft already): $212.6 \times 2 \times 15 = \$6,378$. It means that for a 1500 sq ft house an additional 100 sq ft would increase the price by \$6,378. This is compared to \$9,182 in the linear model in part (b).

- (d) It is hard to tell from the graph. The quadratic model appears to be a little bit better since it captures the curvature of the scatter plot. The SSE is 2.227×10^{12} for the linear model and 2.032×10^{12} for the quadratic model. The quadratic model appears to have smaller SSE.



- (e) For houses on a large lot, $\widehat{price} = 113279.4 + 193.8 \text{ livarea}^2$. For houses on a smaller lot, $\widehat{price} = 62172.4 + 186.9 \text{ livarea}^2$. The two marginal effects (extra 100 sq ft for a 1500 sq ft house) are $193.8 \times 2 \times 15 = \dots$ and $186.9 \times 2 \times 15 = \dots$. Their interpretations are ...
- (f) (I omitted this question – you obviously should not.)
- (g) (I omitted this question – you obviously should not.)